Geometry and Applications

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Introduction

1 Review

1.1 The Radon-Nikodym Theorem

Recall that for a set X, $\mathcal{P}(X)$ denotes its *power set*, i.e. the set of subsets of X.

Definition 1. *Let X be a set and let $\mathcal{F} \subset \mathcal{P}(X)$ be a collection of subsets of X. We say $\mathcal{F}$ is a σ-*algebra *if*

1. *$X \in \mathcal{F}$*
2. *If $A \in \mathcal{F}$ then $X \setminus A \in \mathcal{F}$*
3. *If $A_1, A_2, \ldots \in \mathcal{F}$ is a countable set of members of $\mathcal{F}$ then $A_1 \cup A_2 \cup \cdots \in \mathcal{F}$.*

Definition 2. *A measure space is a triple $(X, \mathcal{F}, \mu)$ where X is a set, $\mathcal{F}$ is σ-algebra of subsets of X, and $\mu : \mathcal{F} \to [0, \infty)$ is a function that satisfies*

1. *$\mu(\emptyset) = 0$*
2. *$\mu(A) \geq 0$ for all $A \in \mathcal{F}$*
3. *If $\{A_1, A_2, \ldots\}$ is a countable set of members of $\mathcal{F}$ such that $A_i \cap A_j = \emptyset$ for $i \neq j$, then $\mu(\cup A_k) = \sum_k \mu(A_k)$.*

We call μ a measure and the sets $A \in \mathcal{F}$ measurable sets. *We often say $(X, \mathcal{F})$ is the* measurable space *while (X, μ) is the* measure space*, as we can have more than one measure.*

Definition 3. *Let $(X, \mathcal{F})$ and $(X', \mathcal{F}')$ be measure spaces and let $f : X \to X'$ be a function. We say f is* measureable *if for every $Y \in \mathcal{F}'$, $f^{-1}(Y) \in \mathcal{F}$.*

Definition 4. *Let (X, μ) be a measure space. We say a function $f : X \to \mathbb{R}$ is* integrable *if $\int_X f d\mu < \infty$.*

Definition 5. *Let μ and ν be two measures on a measurable space $(X, \mathcal{F})$. Then we say ν is absolutely continuous with respect to μ, denoted by $\nu \ll \mu$, if $\mu(A) = 0$ implies $\nu(A) = 0$ for $A \in \mathcal{F}$.*

Theorem 1. *(Radon-Nikodym) Let $(X, \mathcal{F})$ be a measureable space and let μ and ν be measures on $(X, \mathcal{F})$ such that $\nu \ll \mu$. Then there exists a measurable function $f : X \to [0, \infty)$ such that*

$$\nu(A) = \int_X f d\mu \qquad \text{for all } A \in \mathcal{F}.$$

Moreover, this function f is a.e. unique, and we denote it by $f = d\mu/d\nu$. It is called the Radon-Nikodym derivative.

Proof. Let C be the set of measurable functions $g : X \to [0, \infty)$ such that $\int_A g d\mu \leq \eta(A)$ for all $A \in \mathcal{F}$. Now let $\alpha = \sup_{g \in C} \int_X g d\mu$ and let $g_n \in C$ be a sequence of functions such that $\lim_{n \to \infty} \int_A g_n d\mu = \alpha$. Then define $f = \sup_n g_n$. By the monotone convergence theorem, we know $\int_X f d\mu = \alpha$. As C is closed, we have $f \in C$.

We know that $\nu(A) \geq \int_A f d\mu$ because $f \in C$. Let us show that we have equality. Indeed, suppose $\nu(A) > \int_A f d\mu$. [TODO] Therefore, $\nu(A) = \int_A f d\mu$.

To show the uniqueness of f, suppose there were another $\tilde{f}$ which satisfies the condition. Then we would have $\int_A (f - \tilde{f}) d\mu = 0$ for all $A \in \mathcal{F}$, which imples $f = \tilde{f}$ almost everywhere. □

Example 1. *(Fluid Mechanics) Let $R \subset \mathbb{R}^3$ be a region occupied by a fluid. Let λ be the Lebesgue measure, and let μ be a measure where $\mu(A)$ represents the mass of the fluid contained in a measurable subset $A \subset R$. Assume $\mu \ll \lambda$. Then by the Radon–Nikodym theorem, there exists a measurable function $\rho : R \to [0, \infty)$ such that*

$$\mu(A) = \int_A \rho \, d\lambda \quad \text{for all measurable } A \subset R.$$

The function ρ is called the mass density of the fluid.

Example 2. *(Finance)*

1.2 Geometric Spaces

The goal of this section is to introduce the intuitive idea of manifolds and orientation.

Recall the implicit function theorem from multivariable calculus.

Proposition 1. *Let $f : \mathbb{R}^3 \to \mathbb{R}^1$ be a continuously differentiable function, and let $v \in \mathbb{R}^3$. If $\nabla f(v) \neq 0$, then there is a neighborhood U of v, and a diffeomorphism $\phi : U \cong \mathbb{R}^2$ such that $z = \phi(x, y)$ for all $(x, y, z) \in U$.*

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a C^1 function. Fix a $c \in \mathbb{R}$ and let M be the level-set defined by

$$M = \{x \in \mathbb{R}^3 : f(x) = c\}.$$

Assume $\nabla f(v) \neq 0$ for all $v \in M$. The implicit function theorem implies that for each $v \in M$, there is a neighborhood U of v that is diffeomorphic to an open subset of $\mathbb{R}^2$, i.e. it is *locally Euclidean.* We say such spaces are *manifolds.*

Since $\nabla f(v)$ is defined and non-zero for every $v \in \mathbb{R}^3$, we can look at the *tangent space* at each v. We define this as

$$T_v M = \{w \in \mathbb{R}^3 : \nabla f(v) \cdot w = 0\}.$$

Now we can define an *normal* function, which assigns a vector which is perpendicular to the surface at each point. We define it as

$$\mathbf{n} : M \to \mathbb{R}^3 \qquad n(v) = \nabla f(v).$$

We say M is *orientable* if $\mathbf{n}$ is continuous and nowhere vanishing.

Remark 1. *To give another way of thinking about orientation, consider a manifold M, and let $v \in M$ be a point. Then let $\{e_1, e_2\}$ be a basis of $T_v M$. Let $w = e_1 \times e_2$ be the vector perpendicular to both e_1 and e_2. Then we can consider the quantity $\kappa(v) = \det(e_1\ e_2\ w)$ to be the* local orientation *at v. One can apply this notion to the unit sphere, for example, to show that it has a consistent orientation of $\kappa = 1$.*

Example 3. *As an example of a non-orientable space, consider the* mobius band *M. Choose a starting point x_0 on the center of the band, and continuously follow this line with a normal vector until we get back to x_0. When we start, the normal vector will be pointing in one specified direction, and when it ends, it will be in another. Which implies that there is a discontinuous jump of the normal vector.*

1.3 Fundamental Theorems of Vector Calculus

We use the differentials forms as follows

Form	Purpose
r and $d\mathbf{r}$	particle position, integrating along curves
$dA = dx \cdot dy$	area of infinitesimal parallelogram
$dV = dx \cdot dy \cdot dz$	volume of infinitesimal parallelepiped

Definition 6. *Let $\mathfrak{X}$ denote the space of vector fields on $\mathbb{R}^3$. We define the gradient operator $\nabla : \mathfrak{X} \to \mathfrak{X}$ by*

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right).$$

We define the curl *of F by*

$$Curl\ F := \nabla \times F$$

and its divergence by

$$Div\ F = \nabla \cdot F.$$

Proposition 2. *(Stoke's Theorem) Let $S \subset \mathbb{R}^3$ be an oriented surface with boundary ∂S. Let $\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^3$ be a C^1 vector field. Then*

$$\int_{\partial S} \mathbf{F} \cdot dr = \int_S (Curl\ F) \cdot \mathbf{n} dA$$

We have the following special case.

Corolary 1. *(Green's Theorem) Let $D \subset \mathbb{R}^2$ be an oriented region. Let $\mathbf{F} = (P, Q) : \mathbb{R}^2 \to \mathbb{R}^2$ be a vector field. Then*

$$\int_{\partial D} \mathbf{F} \cdot d\mathbf{r} = \int_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Proposition 3. *(Divergence Theorem) Let $\Omega \subset \mathbb{R}^3$ be an oriented volume with boundary $\partial\Omega$. Let $\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^3$ be a vector field. Then*

$$\int_M \mathbf{F} \cdot \mathbf{n} dA = \int_{\partial M} \nabla \cdot \mathbf{F} dV.$$

1.4 Calculus of Variations

We introduce a basic theorem in the Calculus of Variations called the Euler-Lagrange equations.

Lemma 1. *Let $f : [a, b] \to \mathbb{R}$ be an integrable function. Then if*

$$\int_a^b f(t)g(t)dt = 0$$

for all integrable functions $g : [a, b] \to \mathbb{R}$, then $f(t) = 0$ for all $a \leq t \leq b$.

Proof. (Sketch) If $f \neq 0$ then we can assume there is a point $t_0 \in [0, 1]$ such that $f(t_0) \neq 0$, and in fact, we can assume $f(t_0) > 0$. Then we can let g be a "bump function" which is positive in a neighborhood of t_0 and 0 outside. Such a function makes the integral positive, which is a contradiction. □

Definition 7. *(Lagrangian and Action Functional) Let $U \subset \mathbb{R}^n$ be open and let $a < b$. Let $L : [a, b] \times U \times \mathbb{R}^n \to \mathbb{R}$ be a C^1-function. We define the* action functional

$$\mathcal{A} : C^1([a, b]) \to \mathbb{R} \qquad \mathcal{A}(x) = \int_a^b L(t, x(t), x'(t))dt.$$

Definition 8. *A path $x \in C^1([a, b])$ is a stationary point of $\mathcal{A}$ if for every $\eta \in C^\infty([a, b])$ with $\eta(a) = \eta(b) = 0$, we have*

$$\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} \mathcal{A}(x + \varepsilon\eta) = 0.$$

Theorem 2. *If x is stationary for $\mathcal{A}$, then for every $1 \leq i \leq n$, we have*

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}_i} - \frac{\partial L}{\partial x_i} = 0.$$

Proof. Let $\varepsilon > 0$ and let $\eta = (\eta_1, \ldots, \eta_n)$ be a smooth function such that $\eta(a) = \eta(b) = 0$. Let $x_\varepsilon = x + \varepsilon\eta$ be a variation that we will work with. Then as x is stationary, we have

$$\left.\frac{d\mathcal{A}(x_\varepsilon)}{d\varepsilon}\right|_{\varepsilon=0} = 0.$$

The proof consists of expanding this statement and applying the first lemma of this section. To simplify notation, let $u = (t, x + \varepsilon\eta, \dot{x} + \varepsilon\dot{\eta})$. We will differentiation under the integral sign. Thus we use the chain rule and find

$$\left.\frac{d}{dt}\right|_{\varepsilon=0} L(u) = \sum_{i=1}^n \frac{\partial L}{\partial x_i}(u)\eta_i(t) + \sum_{i=1}^n \frac{\partial L}{\partial \dot{x}_i}(u)\dot{\eta}_i(t) = 0.$$

Therefore, if $\phi : (-1, 1) \to \mathbb{R}$ is given by $\phi(\zeta) = A(x_\zeta)$ then the above argument shows that

$$\phi'(0) = \int_a^b \sum_{i=1}^n \frac{\partial L}{\partial x_i}(t, x, \dot{x})\eta_i(t) + \sum_{i=1}^n \frac{\partial L}{\partial \dot{x}_i}(t, x, \dot{x})\dot{\eta}_i(t)dt = 0.$$

Let us write this in the more compact form

$$\phi'(0) = \int_a^b D_x L(t, x, \dot{x})\eta(t) + D_{\dot{x}} L(t, x, \dot{x})\eta'(t)dt$$

and consider the second integrand, which we integrate by parts. Indeed, we have

$$\int_a^b D_{\dot{x}} L(t, x, \dot{x})\eta'(t)dt = D_{\dot{x}} L(t, x, \dot{x})\eta(t)|_a^b - \int_a^b \frac{d}{dt} D_{\dot{x}}(t, x, \dot{x})\eta(t)dt.$$

As $\eta(a) = \eta(b) = 0$, the first term vanishes and we get

$$0 = \phi'(0) = \int_a^b (D_x L(t, x, \dot{x}) - \frac{d}{dt} D_{\dot{x}}(t, x, \dot{x}))\eta(t)dt.$$

As this holds for all η, our first lemma imples our claim. □

Proposition 4. *Let $\tilde{L}$ be a function such that $\tilde{L}(x, \dot{x}, t) = L(x, \dot{x}, t) + \frac{d}{dt}F(x, t)$. Then*

$$\frac{d}{dt}\frac{\partial \tilde{L}}{\partial \dot{x}_i} - \frac{\partial \tilde{L}}{\partial x_i} = 0.$$

Proof. Let S denote the action of L and let S' denote the action of $\tilde{L}$. Then

$$S[q] = S'[q] + (F(q(t_1), t_1) - F(q(t_0), t_0))$$

and we notice that $F(q(t_1), t_1) - F(q(t_0), t_0)$ is a constant. Therefore, a path minimizes S if and only if it minimizes S'. □

1.5 Newton's Laws

Definition 9. *(Newtonian Particle) We begin by introducing the notion of a particle. We work in $Q = \mathbb{R}^3$, which we call the* configuration space. *A particle is defined by its* mass, *which is given by a positive real number m, and a C^2 curve $x : \mathbb{R} \to Q$. We define its* velocity *and* acceleration *by*

$$v = \frac{dx}{dt} = \dot{x} \quad \text{and} \quad a = \frac{d^2x}{dt^2} = \ddot{x}.$$

We define a force field *to be a function $F : Q \times \mathbb{R} \to \mathbb{R}$, which satisfies the following axioms:*

1. *If $F(x(t), t) = 0$ for all t, then $\ddot{x} = 0$ (law of inertia).*
2. *$F(x(t), t) = m\ddot{x}$.*

3. *If x_1 and x_2 are two distinct particles such that x_1 acts on x_2 by force $F_{1,2}$ and x_2 acts on x_1 by force $F_{2,1}$, then $F_{1,2} = -F_{2,1}$ (action-reaction).*

If we have a system of N particles $x_1, \dots, x_n$ and for each i, j there exists a scalar a_{ij} such that $F_{ij} = a_{ij}(x_i - x_j)$, then we say the forces are central.

Proposition 5. *If the forces in the system are central, then*

$$F(\mathbf{x} + a, \dot{\mathbf{x}}) = F(\mathbf{x}, \dot{\mathbf{x}})$$

for all $a \in \mathbb{R}$.

Proof. Write F as $F = (F_1, \dots, F_n)$ where each $F_i : \mathbb{R}^3 \to \mathbb{R}^3$. Then for each i we have

$$F_i = \sum_{j \neq i} F_{ij} = \sum_{j \neq i} a_{ij}(x_i - x_j)$$

for some constants a_{ij}. We have the identity

$$F_i(x_1 + a, \dots, x_n + a) = F_i(x_1, \dots, x_n)$$

which proves our claim. □

We define the *momentum* of a particle x with mass m by $p = m\dot{x}$.

Proposition 6. *(Conservation of momentum) Let $\{x_1, \dots, x_N\}$ be a system of N particles, where particle x_i has mass m_i. Then the time-derivative of the total-momentum is 0, i.e.*

$$\frac{d}{dt} \sum_{i=1}^{N} m_i \dot{x}_i = 0.$$

Proof. This follows directly from Newton's third law. Indeed,

$$\frac{d}{dt} \sum_{i=1}^{N} m_i \dot{x}_i = \sum_{i=1}^{n} F_i$$

where F_i is the total force acted on particle i by the other particles. Thus

$$\sum_{i=1}^{n} F_i = \sum_{i=1}^{N} \sum_{j=1, j \neq i}^{N} F_{ij}.$$

But $F_{ij} = -F_{ji}$ when $i \neq j$ so the sum is 0, proving our claim. □

angular momentum of a particle x with mass m as

$$\ell = x \times (m\dot{x})$$

where $\times$ is the cross-product. We have a version of the product rule for the cross-product, which states that for functions $\mathbf{f}, \mathbf{g} : \mathbb{R} \to \mathbb{R}^3$ we have

$$\frac{d}{dt}(\mathbf{f} \times \mathbf{g}) = \dot{\mathbf{f}} \times \mathbf{g} + \mathbf{f} \times \dot{\mathbf{g}}.$$

Applying this to ℓ gives us

$$\frac{d\ell}{dt} = m(\dot{x} \times \dot{x} + x \times \ddot{x}) = m(x \times \ddot{x}).$$

Remark 2. *If x_1 and x_2 are two particles with a central force F between them, then $(x_1 - x_2) \times F = 0$.*

Proposition 7. *(Conservation of angular momentum) Consider the same system $\{x_1, \dots, x_N\}$ as in the previous proposition, with the extra condition that each pairwise force $F_{i,j}$ between x_i and x_j is central. Then the time-derivative of the total angular momentum is 0, i.e.*

$$\frac{d}{dt} \sum_{i=1}^{N} (x_i \times (m_i \dot{x}_i)) = 0.$$

Proof. By the calculation above, we know

$$\frac{d}{dt} \sum_{i=1}^{N} (x_i \times (m_i \dot{x}_i)) = \sum_{i=1}^{N} m_i (x_i \times \ddot{x}_i).$$

By the bilinearity of the cross product, we get

$$\sum_{i=1}^{N} m_i (x_i \times \ddot{x}_i) = \sum_{i=1}^{N} \sum_{j \neq i} (x_i - x_j) \times F_{i,j}.$$

This sum is zero by the centrality of the forces $F_{i,j}$. □

Definition 10. *Assume we are working on a region R. We say a force F is* conservative *if $F = -\nabla V$ for some function V. This is equivalent to Curl $F = \nabla \times F = 0$.*

Definition 11. *We define the* kinetic energy *of a particle as*

$$T = \frac{1}{2} m \dot{x}^2.$$

We define its total energy *by*

$$E = T + V.$$

Proposition 8. *Let $\{x_1, \ldots, x_N\}$ be a system of N particles with potential V. Then the time-derivative of the total energy is 0, i.e.*

$$\frac{d}{dt} \sum_{i=1}^{n} (\frac{1}{2} m_i \dot{x_i}^2) + V = 0.$$

Proof. We notice

$$\frac{d}{dt} \sum_{i=1}^{n} \frac{1}{2} m_i \dot{x_i}^2 = \sum_{i=1}^{N} m_i \ddot{x} = F$$

so the sum in the proposition is 0 by the definition of the potential. □

1.6 The Lagrangian

Definition 12. *For a Newtonian system with Kinetic energy T and potential V, we define the* (canonical) Lagrangian *as*

$$L : \mathbb{R}^{3N} \times \mathbb{R}^{3N} \times \mathbb{R} \to \mathbb{R} \qquad L = T - V.$$

For a Newtonian system with Kinetic energy

Explicitly, with coordinates, we plug in the positions, velocities and then time, i.e. we look at the quantity

$$L(x, \dot{x}, t) = \frac{1}{2} \sum_{i=1}^{N} m_i \dot{x}^2 + V(x(t)).$$

Definition 13. *For a path $q : [t_0, t_1] \to \mathbb{R}^3$, we define the* action *by*

$$S[q] := \int_{t_0}^{t_1} L(q, \dot{q}, t) dt.$$

In the manifold setting, we consider L as a function $L : TQ \to \mathbb{R}$. In this setting, we consider L with the notation $L(x_1, \ldots, x_n, \dot{x_1}, \ldots, \dot{x_n})$.

Proposition 9. *(Euler-Lagrange) For a Newtonian system with N particles we have*

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x_i}} - \frac{\partial L}{\partial x_i} = 0.$$

Proof. This is a special case of the theorem we proved in the section on Calculus of Variations. □

Proposition 10. *If L is the Lagrangian of our system and $\tilde{L}(x, \dot{x}, t) = L(x, \dot{x}, t) + \frac{d}{dt} G(x, t)$ defines the same equations of motion for any smooth function G.*

Proof. See the section on Calculus of Variations. □

1.7 Hamiltonian Mechanics

Let $L = L(q, \dot{q}, t)$ be the canonical Lagrangian. Define

$$p_i = \frac{\partial L}{\partial \dot{q_i}}.$$

If the Hessian of L with respect to $\dot{q}$ has non-zero determinant, i.e.

$$\det \frac{\partial L}{\partial \dot{q_i} \dot{q_j}} \neq 0$$

then the inverse function theorem tells us that we can solve for $\dot{q}$ in terms of q and p, i.e. $\dot{q} = q(p, q)$. We then define the *Hamiltionian (with respect to L)* as

$$H(q, p, t) = p \cdot \dot{q} - L(q, \dot{q}, t).$$

We then find

$$\dot{q_i} = \frac{\partial H}{\partial p_i} \qquad \dot{p_i} = -\frac{\partial H}{\partial q_i}.$$

Manifolds

2 Smooth Manifolds

Definition 14. *A n-dimensional* topological manifold *M is a second countable, Hausdorff toplogical space such that for every $x \in M$ there is a neighborhood U of x such that U is homeomorphic to $\mathbb{R}^n$. A* k-smooth atlas *on M is a collection $\{(U_\alpha, \phi_\alpha)\}$ such that (a) $\{U_\alpha\}$ covers M and for each α, β, the function $\varphi_\beta \circ \varphi_\alpha^{-1}$ restricted to $\varphi(U_\alpha \cap U_\beta) \to \mathbb{R}^n$ is C^k. An* n-dimensional C^k manifold *is an n-dimensional topological manifold equipped with a C^k-smooth atlas. We say M is a **smooth manfiold** if $k = \infty$.*

Proposition 11. *If M is a manifold, then M has a countable atlas.*

Proof. See [Spivak 1]. □

Let M and N be smooth manifolds of dimensions m and n, respectively, and let $f : M \to N$ be a function. Let $x_0 \in M$. Let (U, φ) be a chart containing x_0 and let (V, ψ) be a chart containing $f(x_0)$. Then the function $\psi \circ f \circ \varphi^{-1}$ is a function from $\varphi(U) \subset \mathbb{R}^m$ to $\psi(V) \subset \mathbb{R}^n$, and we say f is C^k*-smooth* at x_0 if the map $\psi \circ f \circ \varphi^{-1}$ is smooth at $\varphi(x_0)$. It is worth noting that this definition is independent of the charts chosen. Indeed, if (U', α) and (V', β) are two other charts of x_0 and $f(x_0)$, respectively, then by the fact that all charts are diffeomorphism, we get that $\beta \circ f \circ \alpha^{-1}$ is also smooth.

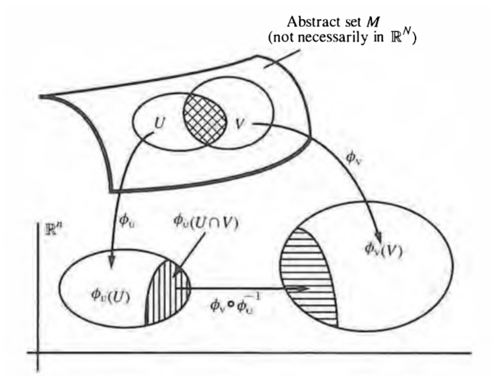

Figure 1: Manifold Charts

If (U, ϕ) is a coordinate chart on M, then there exist smooth functions $\phi_1, \ldots, \phi_n : M \to \mathbb{R}$ such that for every $x \in M$, we have $\phi(x) = (\phi_1(x), \ldots, \phi_n(x))$. We call the ϕ_i's local-coordinates, and denote them by $x^i = \phi_i(x)$.

Definition 15. *A collection of functions $\phi_\alpha : M \to [0, 1]$ is called a* partition of unity *if for every $x \in M$ there is a open set U containing x such that $(\phi_\alpha)|_W = 0$ for all but finitely many α and $\sum_\alpha \phi_\alpha = 1$. We say it is subordinate to an open cover U_α if support$(\phi_\alpha) := \overline{\{x \in M : \phi_\alpha(x) \neq 0\}} \subset U_\alpha$ for every α.*

Proposition 12. *M has a countable atlas $\{U_1, \ldots\}$ and a partition of unity $\{\phi_1, \ldots\}$ subordinate to the atlas.*

2.1 Tangent vectors

In this section we introduce the notion of a tangent vector.

Definition 16. *Let $x_0 \in M$. We say two curves $\gamma_1, \gamma_2 : (-1, 1) \to M$ which satisfy $\gamma_1(0) = \gamma_2(0) = x_0$ are equivalent if for every $f \in C^\infty(M)$, we have*

$$\frac{d(f \circ \gamma_1)}{dt}(0) = \frac{d(f \circ \gamma_2)}{dt}(0).$$

At tangent vector to x_0 *is such an equivalence class and we denote all such equivalence classes by $T_p M$, which we call the* tangent space at x_0.

We recall that a *derivation of M at $p \in M$* is an $\mathbb{R}$-linear map $D : C^\infty(M) \to \mathbb{R}$ such that for all $f, g \in C^\infty(M)$ we have $D(fg) = D(f)g(p) + f(p)D(g)$. We denote the vector space of derivations by $\mathrm{Der}_p(C^\infty(M))$.

Proposition 13. *Let $(U, \phi) = (U, x^1, \ldots, x^n)$ be a coordinate chart around p. For $1 \leq i \leq n$, let $\partial_i = \dfrac{\partial}{\partial x^i}(p)$. Then $\{\partial_1, \ldots, \partial_n\}$ is a basis of $Der_p(C^\infty(M))$.*

Proof. We first show that $\{\partial_1, \ldots, \partial_n\}$ are linearly indepdendent. Indeed, if $a^i \partial_i = 0$, then for $1 \leq j \leq n$, we can define $f_j : M \to \mathbb{R}$ by $f = x^j$. Then $a^i \partial_i f_j = a^j = 0$. Thus $\partial_1, \ldots, \partial_n$ are linearly indepdendent.

To show that they span $\mathrm{Der}_p(C^\infty(M))$, let $f \in C^\infty(M)$ and consider its Taylor expansion around p,

$$f(x) = f(p) + \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(x^i - p^i) + R(x)$$

where $R(x)$ consists of higher order terms. Now let $D \in \mathrm{Der}_p(C^\infty(M))$. Then as $D(\text{const}) = 0$ we get

$$D(f) = \sum_{i=1}^{n} a^i \frac{\partial f}{\partial x^i} \qquad a^i = D(x^i).$$

So $\partial_1, \ldots, \partial_n$ span $\mathrm{Der}_p(C^\infty(M))$ as claimed. □

For a tangent vector $[\gamma] \in T_pM$, we define its derivation by

$$D_\gamma(f) = \frac{d(f \circ \gamma)}{dt}(0) \qquad f \in C^\infty(M).$$

Corolary 2. *The map*

$$T_pM \to Der_p(C^\infty(M)) \qquad [\gamma] \mapsto D_\gamma$$

is a vector space isomorphism.

Proof. The map is injective by the definition of the tangent space. The map is surjective by the proof of the above theorem. □

2.2 Bundles

Definition 17. *A* fiber bundle *with fiber F consists of a continuous surjection $\pi : E \to M$ such that for every $x \in M$ there exists an open set U containing x and a homeomorphism $\rho : \pi^{-1}(U) \to U \times F$ such that $\pi = pr_U \circ \rho$.*

Definition 18. *If F is a real finite dimensional vector space, we say π is a* vector bundle.

Definition 19. *A section of a bundle $\pi : E \to M$ is a smooth map $s : M \to E$ such that $\pi \circ s = id_M$. We denote the space of sections of E by $\Gamma(E)$.*

Definition 20. *Given an n-dimensional smooth manifold M, we define its* tangent bundle *TM as*

$$TM = \bigsqcup_{p \in M} T_pM$$

together with a projection map $\pi : TM \to M$ given by $\pi(v_p) = p$. We define a manifold structure on TM as follows. Let (U, φ) be a chart in M. Then for every $p \in U$, we can write $v_p = \sum v^i \frac{\partial}{\partial x^i}(p)$. Then we can define a chart $(\pi^{-1}(M), \tilde{\varphi})$ by

$$\tilde{\varphi}(v_p) = (\varphi(p), (v^1, \ldots, v^n)).$$

This makes TM into a $2n$-dimensional manifold.

Definition 21. *We define the cotangent bundle as*

$$T^*M = \bigsqcup_{p \in M} T_pM^*$$

Definition 22. *For $r, s \geq 0$, define the (r, s) tensor bundle as*

$$T^{r,s}M = TM^{r\otimes} \otimes T^*M^{s\otimes}.$$

Definition 23. *A vector field X is a member of $\Gamma(TM)$.*

Definition 24. *A k-*form *is a member of $\Gamma(\Lambda^k T^*M)$. We also denote this by $\Omega^k(M)$.*

Vector Bundle	Section	Section Notation
Tangent Bundle TM	Vector Fields	$\mathfrak{X}(M)$
Alternating Bundle $\Lambda^k TM$	k-forms	$\Omega^k(M)$
Tensor Bundle $T^{r,s}M$	Tensors	$\mathfrak{T}^{r,s}(X)$

2.3 Basics of Vector Fields

We outline the basic theory of vector fields.

Lemma 2. *(Picard-Lindelof) Let $U \subset \mathbb{R}^n$ be open, and let $f : U \to \mathbb{R}$ be a C^1 function. For any $x_0 \in U$, the initial value problem*

$$\frac{dx}{dt} = f(x) \qquad x(0) = x_0$$

has a unique smooth solution $x : (-1, 1) \to U$.

Proposition 14. *Let X be a vector field on M and let $p_0 \in M$. Then there is a unique solution $\Phi_t^X(p_0) : (-1, 1) \to M$ of the differential equation*

$$\frac{d\Phi_t^X(p_0)}{dt} = X(\Phi_t^X(p_0)).$$

Proof. This is immediate from the Picard-Lindelof theorem. □

2.4 Differential Forms and the Exterior Derivative

Recall that a k-form is a member of $\Gamma(\Lambda^k T^*M)$. But if V is a vector space, then it is common to identify $\Lambda^1 V$ with its dual V^*. Hence we view one-forms as members of $\Gamma(T^*M) = \Omega^1(M)$.

Definition 25. *Fix $p \in M$ and let $\{\frac{\partial}{\partial x^i}(p) : 1 \leq i \leq n\}$ be a basis of T_pM. We define its dual basis, i.e. the basis of T_pM^*, as $\{(dx^i)_p : 1 \leq i \leq n\}$, i.e.*

$$(dx^i)_p \frac{\partial}{\partial x^j}(p) = \delta_{ij}.$$

As we can define $(dx^i)_p$ for each $p \in M$, each dx^i is a legimate one-form, i.e. a section of T^*M.

Definition 26. *Let $f : M \to \mathbb{R}$ be smooth and fix $p \in M$. Then we define the linear map $(df)_p : T_pM \to \mathbb{R}$ by*

$$(df)_p(X_p) = X_p(f).$$

Again, df is a section of T^*M.

Proposition 15.

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i.$$

Proof. Suppose $X = a^i \partial_i$. Then $(df)(X) = X(f) = a^i \partial_i(f)$. Moreover, $\partial^i(f) dx^i X = a^i \partial_i(f)$ which proves the statement. □

Proposition 16. *Every k-form $\omega \in \Omega^k(M)$ has the form*

$$\omega = \sum \omega_{i_1,\dots,i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k}$$

where $\omega_{i_1,\dots,i_k} \in C^\infty(M)$.

Proof. Recall that members of $\Omega^k(M)$ are functions $\omega : M \to \Lambda^k TM$. So if $p \in M$, then $\omega(p)$ is a multilinear map $(T_pM)^k \to \mathbb{R}$. Now if $\frac{\partial}{\partial x^i}(p)$, $1 \leq i \leq n$ be a basis of T_pM, and let dx^i, $1 \leq i \leq n$ be its dual basis. By multilinear algebra, we know $\omega(p)$ has the form

$$\omega(p) = \sum_{1 \leq i_1 < \cdots < i_k \leq n} \omega_{i_1,\dots,i_k}(p) dx^{i_1} \wedge \cdots dx^{i_k}$$

where the sum ranges over all subsets of $\{1, \dots, n\}$ of cardinality k, and $\omega_{i_1,\dots,i_k} \in C^\infty(M)$ for each $\{i_1, \dots, i_k\}$. □

Remark 3. *We can evaluate k-forms as follows. Consider the simple k-form*

$$f dx^{i_1} \wedge \cdots \wedge dx^{i_k} \qquad f \in C^\infty(M).$$

Then if $X_1, \dots, X_k \in \mathfrak{X}(M)$, then

$$(f dx^{i_1} \wedge \cdots \wedge dx^{i_k})(X_1, \dots, X_k) = f \cdot \det \begin{pmatrix} (dx^{i_1})(X_1) & \cdots & (dx^{i_1})(X_k) \\ \vdots & \cdots & \vdots \\ (dx^{i_k})(X_1) & \cdots & (dx^{i_k})(X_k) \end{pmatrix} = f \det(x_{ij}) \qquad X_i = (x_1, \dots, x_n).$$

We are now ready to define the differential of a function, which is the analog of the Jacobian for manifolds.

Definition 27. *Let $f : M \to N$ be a smooth map of manifolds, and fix a $p \in M$. Let v_a be a member of T_pM. Then if $g \in C^\infty(N)$ then we can define the* pushforward *or* differential *$f_{*,p} : T_pM \to T_{f(p)}N$ by*

$$f_{*,p}(X)(g) := X(g \circ f).$$

As we have this for each p, we get a map between tangent bundles

$$f_* : TM \to TN$$

which restricts to $f_{*,p}$ on each T_pM. We also denote f_* by df.

Proposition 17. *f is smooth if and only if f_* is smooth.*

Proof. ... □

Remark 4. *If $f \in C^\infty(M)$ and $\alpha \in \Omega^k(M)$, then we define $f \wedge \alpha = f \cdot \alpha$ (usual multiplication). This makes the below proposition true.*

We now extend d from $C^\infty(M)$ to $\Omega^\bullet(M)$.

Proposition 18. *For each $k \geq 0$, there is a unique derivation $d : \Omega^k(M) \to \Omega^{k+1}(M)$ such that*

1. *d is linear*
2. *$d^2 = d \circ d = 0$*

3. *If $\omega \in \Omega^k(M)$ and $\eta \in \Omega^l(M)$ then $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$*

Proof. We already have a map $d : C^\infty(M) \to \Omega^1(M)$. Let us use rule (3) to expand such a d on k-forms. Indeed, for a set $I \subset \{1, \dots, n\}$ of cardinality k and a function $f \in C^\infty(M)$, we can use rules (1), (2) and (3) together to get

$$d(f dx^I) = df \wedge dx^I = \sum_{i \notin I} \frac{\partial f}{\partial x^i} dx^i \wedge dx^I.$$

Which completes the proof. □

Definition 28. *The above definition of d gives us the co-chain complex*

$$\cdots \to \Omega^k(M) \xrightarrow{d_k} \Omega^{k+1}(M) \xrightarrow{d_{k+1}} \cdots$$

from which we obtain the de Rham cohomology groups

$$H^k_{DR}(M) = \ker(d_n)/im\ (d_{n-1}).$$

Proposition 19. *Let $\psi : M \to N$ be a smooth map of manifolds. Then there is a unique graded algebra homomorphism $\psi^* : \Omega(N) \to \Omega(M)$ such that*

1. *For smooth functions $f : N \to \mathbb{R}$, $\psi^* f = f \circ \psi : M \to \mathbb{R}$.*
2. *For one forms $\alpha : T_p M^* \to \mathbb{R}$, we have $\psi^* \alpha = \alpha \circ d\psi$*
3. *$\psi^*(\alpha \wedge \beta) = \psi^*(\alpha) \wedge \psi^*(\beta)$.*

2.5 Oriented Manifolds

Let M be a smooth manifold of dimension n.

Definition 29. *We say an atlas $\{(U_\alpha, \phi_\alpha)\}$ is* oriented *if for all α, β, the Jacobian of*

$$x_\beta \circ x_\alpha^{-1} : \mathbb{R}^n \to U_\alpha \cap U_\beta \to \mathbb{R}^n$$

has positive determinant.

Definition 30. *Let M be an oriented manifold with oriented atlas $\{(U_\alpha, x_\alpha)\}$. A volume form ν is called* positive *if for every chart (U_α, x_α) in the oriented atlas we have*

$$\nu = f\, dx^1_\alpha \wedge \cdots \wedge dx^n_\alpha \quad \text{with } f > 0 \text{ on } U_\alpha.$$

Proposition 20. *If M is oriented, then M has a volume form.*

Proof. Let (U_α, x_α) be an oriented atlas. Let $p \in M$ and let (U_α, x_α) and (U_β, x_β) be charts such that $p \in U_\alpha \cap U_\beta$. To simplify notation, let $x^i = x^i_\alpha$ and $y^i = x^i_\beta$. Note that for each i, we have

$$dx^i = \sum_{j=1}^n \frac{\partial x^i}{\partial y^j} dy^j.$$

Therefore, if

$$\nu_p = dx^1 \wedge \cdots \wedge dx^n$$

then

$$\nu_p = \det \frac{\partial x^\alpha}{\partial y^\beta} dy^1 \wedge \cdots \wedge dy^n.$$

But that coefficient - the determinant of the Jacobian of the transition map - is positive because M is oriented. So if we let $\{\varphi_\alpha\}$ be a partition of unity subordinate to $\{U_\alpha\}$, which we can assume to be countable, then we can define

$$\nu = \sum_\alpha \varphi_\alpha \nu_\alpha$$

which is a volume form. □

Proposition 21. *If M has a volume form, then M is oriented. Moreover, in the oriented chart system, the volume form is positive.*

Proof. Let ν be a volume form and let $p \in M$. Let $\{(U_\alpha, x_\alpha)\}$ be a set of charts of M such that for each α, $\nu_\alpha = f dx^1_\alpha \wedge \cdots \wedge dx^n_\alpha$ and $f > 0$ on U_α. Notice that while $f \neq 0$ is guaranteed, we can assume $f > 0$ since if $f < 0$, then one can take $y^1 = x^2$, $y^2 = x^1$, $y^i = x^i$ for $i > 2$, and then

$$\nu_\alpha = -f dy^1 \wedge \cdots \wedge dy^n.$$

Now if $p \in U_\alpha \cap U_\beta$, then by the last argument, we have

$$\nu_p = \det \frac{\partial x^\alpha}{\partial y^\beta} dy^1 \wedge \cdots \wedge dy^n$$

which must be positive because of how we chose the set of charts. □

Corolary 3. *The following are equivalent.*

1. *M is oriented.*
2. *M has a volume form.*
3. *M has a positive volume form.*

2.6 Cartan Calculus and the Poincare Lemma

Definition 31. *Let M be a smooth manifold of dimension n. Let X be a vector field on M and let Y be a tensor field. We define*

$$L_X Y = \left.\frac{d}{dt}\right|_0 \Phi_t^{X*} Y.$$

Definition 32. *Let $X \in \mathfrak{X}(M)$. We define the map*

$$\iota_X : \Omega^k(M) \to \Omega^{k-1}(M) \qquad \omega \in \Omega^k(M) \quad \iota_X(\omega) = \omega(X, -).$$

Lemma 3. *(Straitening theorem for vector fields) Let $X \in \mathfrak{X}(M)$. Then there is a coordinate chart $(U, x^1, \dots, x^n)$ of M such that $X = \dfrac{\partial}{\partial x^1}$ in local coordinates.*

Remark 5. *We wish to analyze $L_X\omega$ where X is a vector field and ω is a k-form. Let $(U, x^1, \dots, x^n)$ be the local coordinates. Let $\omega = f dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ be a simple k-form. We can re-order $x^2, \dots, x^n$ such that ω necessarily has the form*

$$\omega = f dx^1 \wedge \cdots \wedge dx^k$$

or

$$\omega = f dx^2 \wedge \cdots \wedge dx^{k+1}.$$

Proposition 22. *(Cartan's magic formula) Let X be a vector field and let ω be a k-form. Then*

$$L_X\omega = d\iota_X\omega + \iota_X d\omega.$$

Proof. Choose local coordinates $x^1, \dots, x^n$ such that $X = \partial/\partial x^1$. There are two cases.

First, assume $\omega = f dx^1 \wedge \cdots \wedge dx^k$. Then

$$\iota_X\omega = f dx^2 \wedge \cdots \wedge dx^k$$

so

$$d\iota_X\omega = \frac{\partial f}{\partial x^1} dx^1 \wedge \cdots \wedge dx^k + \sum_{i=k+1} \frac{\partial f}{\partial x^i} dx^i \wedge dx^2 \cdots \wedge dx^k.$$

Moreover,

$$d\omega = \sum_{i=k+1}^{n} \frac{\partial f}{\partial x^i} dx^i \wedge dx^1 \wedge \cdots \wedge dx^k$$

so

$$\iota_X d\omega = -\sum_{i=k+1}^{n} dx^i \wedge dx^2 \wedge \cdots \wedge dx^k$$

thus

$$d\iota_X\omega + \iota_X d\omega = \frac{\partial f}{\partial x^1} dx^1 \wedge \cdots \wedge dx^k.$$

In the second case, assume $\omega = f dx^2 \wedge \cdots dx^{k+1}$. Then $\iota_X\omega = 0$. Moreover,

$$d\omega = \frac{\partial f}{\partial x^1} dx^1 \wedge dx^2 \wedge \cdots \wedge dx^{k+1} + \sum_{i=k+2}^{n} \frac{\partial f}{\partial x^i} dx^i \wedge dx^2 \wedge \cdots dx^{k+1}.$$

So

$$\iota_X d\omega = \frac{\partial f}{\partial x^1} dx^2 \wedge \cdots \wedge dx^{k+1}.$$

Thus

$$\iota_X d\omega + d\iota_X\omega = \iota_X d\omega.$$

In both cases, the sum $\iota_X d\omega + d\iota_X\omega = L_X\omega$. □

Proposition 23. *If X and Y are vector fields, then*

$$L_X Y = XY - YX.$$

Proof. Let $p \in M$ and $f \in C^\infty(M)$. Then

$$L_X Y = \lim_{t \to 0} \frac{\Phi_t^{X*} Y - Y}{t}.$$

Notice that

$$(\Phi_t^{X*} Y)(p)(f) = Y(\Phi_t^X(p))(f \circ \Phi_{-t}^X)$$

and define

$$H(s,t) = Y(\Phi_s^X(p))(f \circ \Phi_t^X).$$

We see that

$$L_X Y = \left.\frac{d}{dt}\right|_0 H(t,t) = \partial_x H(0,0) + \partial_y H(0,0).$$

To compute $\partial_x H(0,0)$, we define $g(q) = Y(f)(q)$ so

$$\partial_x H(0,0) = \frac{d}{dt}\bigg|_{t=0} g(\Phi_t(p)) = X(g) = X(Y(f))(p).$$

Similarly, we have

$$\partial_y H(0,0) = \frac{d}{dt} Y(p)(f \circ \Phi_{-t}) = Y(p)(-X(f)) = -Y(X(f))(p).$$

So

$$L_X Y = XY - YX.$$

□

For vector fields X and Y, we use the notation

$$L_X Y = [X, Y].$$

Example 4. *A* Lie group *is a group G such that (a) G is a smooth manifold (b) the multiplcation map $G \times G \to G$ is smooth and (c) the inversion map $G \to G$ is smooth.*

Example 5. *Diff(M) is a Lie group.*

Remark 6. *Let $F : M \to N$ be a smooth map and let $\omega \in \Omega^k(M)$. Then*

$$F^* \omega = \omega \circ dF$$

and moreover

$$(F^*\omega)_x(v_1, \dots, v_n) = \omega_{F(x)}((dF)_p v_1, \dots, (dF)_p v_n).$$

Definition 33. *A set $U \subset \mathbb{R}^n$ is said to be* star shaped *if there exists $x_0 \in U$ such that for every $y \in U$, the line segment $L(x_0, y) = \{tx_0 + (1-t)y : 0 \leq t \leq 1\}$ is a subset of U.*

Proposition 24. *Let $U \subset \mathbb{R}^n$ be an open star-shaped set. Then*

$$H^k_{DR}(U) = \{0\}$$

for all $k \geq 1$.

Proof. Let ω be a k-form such that $d\omega = 0$. Then

$$\mathcal{L}_X \omega = \frac{d}{dt}\bigg|_{t=0} \Phi_t^* \omega$$

and more generally,

$$\frac{d}{dt} \Phi_t^* \omega = \Phi_t^*(\mathcal{L}_X \omega).$$

Therefore,

$$\Phi_1^*\omega - \Phi_0^*\omega = \int_0^1 (\frac{d}{dt}\Phi_t^*\omega)dt = \int_0^1 (\Phi_t^* \mathcal{L}_X \omega)dt.$$

By Cartan's identity, we know

$$\mathcal{L}_X \omega = \iota_X d\omega + d\iota_X \omega$$

so

$$\int_0^1 \Phi_t^*(\iota_X d\omega + d\iota_X\omega)dt = d\left(\int_0^1 (\Phi_t^* \iota_X \omega)dt\right) + \int_0^1 (\Phi_t^* \iota_X d\omega)dt.$$

Now define $K : \Omega^k(U) \to \Omega^{k-1}(U)$ by

$$K\omega = \int_0^1 (\Phi_t^* \iota_X \omega)dt.$$

Then the above equations imply that

$$Kd + dK = \mathrm{id}_{\Omega^k(U)}.$$

Since $d\omega = 0$, we get $dK\omega = \omega$, which proves our claim. □

2.7 Stoke's Theorem

Before proving Stoke's theorem on Manifolds, let us first prove it in $\mathbb{R}^n$. For $n \geq 1$, define $B^n = [0,1]^n$. Notice that

$$\partial B^n = \bigcup_{i=1}^N F_i \qquad F_i \text{ oriented face of } B^n.$$

Lemma 4. *(Stoke's on boxes in $\mathbb{R}^n$) Let $U \subset \mathbb{R}^n$ be an open set containing B^n, and let $\omega \in \Omega^{n-1}(U)$ be an $n-1$ form. Then*

$$\int_{B^n} d\omega = \int_{B^{n-1}} \omega.$$

Proof. First write ω as

$$\omega = \sum_{i=1}^{n} \omega_i \widehat{dx^i}$$

where

$$\widehat{dx^i} = dx^1 \wedge \cdots dx^{i-1} \wedge dx^{i+1} \wedge \cdots \wedge dx^n.$$

Then

$$d\omega = \left(\sum_{i=1}^{n} (-1)^{i-1} \frac{\partial \omega_i}{\partial x^i} \right) dx^1 \wedge \cdots \wedge dx^n.$$

Now we compute and find

$$\int_{B^n} d\omega = \sum_{i=1}^{n} (-1)^{i-1} \int_{B^n} \frac{\partial \omega_i}{\partial x^i} dx^1 \wedge \cdots \wedge dx^n.$$

We notice that

$$\int_{B^n} \frac{\partial \omega_i}{\partial x^i} dx^1 \wedge \cdots \wedge dx^n = (-1)^{i-1} \int_{F_i} \omega_i \widehat{dx^i}.$$

Therefore,

$$\int_{B^n} d\omega = \sum_{i=1}^{n} \int_{F_i} \omega_i \widehat{dx^i} = \int_{\partial B^n} \omega.$$

□

Lemma 5. *(Stoke's Theorem on domains in $\mathbb{R}^n$) Let $U \subset \mathbb{R}^n$ be an open set, and let $D \subset U$ be a compact set with piecewise smooth boundary. Let $\omega \in \Omega^{n-1}(U)$. Then*

$$\int_D d\omega = \int_{\partial D} \omega.$$

Proof. Let $\mathcal{B} = \{B_1, B_2, \ldots, B_N\}$ be a covering of U by boxes and let $\{\phi_1, \ldots, \phi_N\}$ be a partition of unity subordinate to $\mathcal{B}$. Then

$$\begin{aligned}
\int_D d\omega &= \sum_{i=1}^{N} \int_D \phi_i d\omega \\
&= \sum_{i=1}^{N} \int_{\partial D} \phi_i \omega_i \\
&= \sum_{i=1}^{N} \int_{\partial D} \omega_i \\
&= \sum_{i=1}^{n} \int_{\partial D} \omega \\
&= \int_{\partial D} \omega.
\end{aligned}$$

□

Definition 34. *Let $f : M \to \mathbb{R}^m$ be a smooth map. Let (U, ϕ) be a coordinate chart on M. Then $\phi^* f := f \circ \phi^{-1} : \mathbb{R}^m \to M \to \mathbb{R}^n$ is smooth. We call this the* pullback of f by ϕ.

Definition 35. *Let M be a smooth oriented manifold of dimension n. Let $\{(U_k, \psi_k : U_k \xrightarrow{\cong} V_k \subset \mathbb{R}^n)\}$ be a countable atlas of M and let $\{\phi_k : M \to [0,1]\}$ be a partition of unity subordinate to the atlas. Let ω be a compactly supported n-form of M. We define the* integral of ω on M *by*

$$\int_M \omega = \sum_{k=1}^{\infty} \int_{V_k} (\psi_k^{-1})^* \omega$$

Where the inner integral is the Lebesgue integral.

Proposition 25. *The definition of the integral does not depend on the partition of unity chosen.*

Definition 36. *We say M has boundary if each of the chart maps (U, ϕ) are such that $\phi(U) \subseteq \mathbb{H}^n = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_n \geq 0\}$.*

Theorem 3. *(Stoke's Theorem) Let M be a real orientable smooth manifold of dimension n with boundary, and let $\omega \in \Omega^{n-1}(M)$ be a differential form with compact support. Then*

$$\int_M d\omega = \int_{\partial M} \omega.$$

Proof. Let us apply the definition directly. We have

$$\int_M d\omega = \sum_{k=1}^{\infty} \int_{V_k} (\psi_k^{-1})^* d\omega.$$

By Stoke's theorem on $\mathbb{R}^n$, we get

$$\sum_{k=1}^{\infty} \int_{V_k} (\psi_k^{-1})^* d\omega = \sum_{k=1}^{\infty} \int_{\partial V_k} (\psi_k^{-1})^* \omega$$

which we notice is equal to

$$\int_{\partial M} \omega.$$

□

Definition 37. *Let $X \in \mathfrak{X}(M)$ be a vector field, and let $\mu \in \Omega^n(M)$ be a volume form. We define the* divergence *with respect to μ as*

$$div_\mu X := \mathcal{L}_X \mu.$$

Theorem 4. *(Divergence theorem)*

$$\int_M (\mathrm{div}_\mu X)\,\mu = \int_{\partial M} \iota_X \mu.$$

Proof. Cartan's identity tells us that

$$\mathcal{L}_X \mu = (d\iota_X + \iota_X d)\mu$$

so

$$\int_M (d\iota_X + \iota_X d)\mu = \int_{\partial M} \iota_X \mu$$

as the other summand is zero as $d\mu = 0$. □

2.8 Simplcial Theory

It is worth it to organize our ideas relating simplicies, integration and topology, as this will make the section on algebraic topology easier.

3 Algebraic Topology

We begin with a brief and informal discussion of homotopy, followed by basic treatment of homology. While algebraic topology is facinating, our main goal is to develop computational tools that will help with geometric calculations. We let X be a "sufficiently nice" path connected space.

3.1 Category Theory

Recall that we use the notion of a "class" rather than a set when defining a category to get around logical issues regarding cardinality.

Definition 38. *A* category $\mathcal{C}$ *consists of the following*

1. *A class of objects $Ob(\mathcal{C})$*
2. *A class of morphisms $Mor(\mathcal{C})$*
3. *A class function $dom : Mor(\mathcal{C}) \to Ob(\mathcal{C})$*
4. *A class function $cod : Mor(\mathcal{C}) \to Ob(\mathcal{C})$*
5. *For objects A, B, we can consider their class of morphisms $Hom(A, B)$*
6. *There is a composition class function*

$$Hom(A, B) \times Hom(C, A) \to Hom(C, B) \qquad (f, g) \mapsto f \circ g.$$

7. *Composition is associtative, i.e. $f \circ (g \circ h) = (f \circ g) \circ h$.*
8. *For every $A \in Ob(\mathcal{C})$ there is a unique map $id_A : A \to A$ such that for every $B \in Ob(\mathcal{C})$ and every morphism $g : B \to A$, the map $id_A \circ g = g$ and every map $h : A \to B$ we have $h \circ id_A = h$.*

We also call morphisms *arrows.*

Definition 39. *Let $f : A \to B$ be a morphism.*

1. *f is a* monomorphism *if it is left-cancellable, i.e. for every $g_1, g_2 : B \to A$, we have*

$$fg_1 = fg_2 \implies g_1 = g_2.$$

2. f *is an* epimorphism *if it is right-cancellable, i.e. for every* $g_1, g_2 : B \to C$ *we have*

$$g_1 f = g_2 f \implies g_1 = g_2.$$

Remark 7. *Under reasonable conditions, we can assume:*

1. *Monomorphism = left-cancellable = left-invertible = injective*
2. *Epimorphism = right-cancellable = right-invertible = surjective*

Definition 40. *Let* $\mathcal{C}, \mathcal{D}$ *be categories. A* covariant functor $F : \mathcal{C} \to \mathcal{D}$ *associates to each object* $A \in Ob(\mathcal{C})$ *an object* $F(A) \in \mathcal{D}$ *such that for every morphism* $f : A \to B$*, where* $B \in Ob(\mathcal{C})$*, there is a morphism* $F : F(A) \to F(B)$ *such that*

1. $F(id_A) = id_{F(A)}$.
2. *If* $g : B \to C$ *is a morphism then* $F(g \circ f) = F(g) \circ F(f)$.

Definition 41. *Let* $\mathcal{C}, \mathcal{D}$ *be categories. A* contravariant functor $F : \mathcal{C} \to \mathcal{D}$ *associates to each object* $A \in Ob(\mathcal{C})$ *an object* $F(A) \in \mathcal{D}$ *such that for every morphism* $f : A \to B$*, where* $B \in Ob(\mathcal{C})$*, there is a morphism* $F : F(B) \to F(A)$ *such that*

1. $F(id_A) = id_{F(A)}$.
2. *If* $g : B \to C$ *is a morphism then* $F(g \circ f) = F(f) \circ F(g)$.

Definition 42. *Let* $\mathcal{C}$ *and* $\mathcal{D}$ *be categories. Let* $F, G : \mathcal{C} \to \mathcal{D}$ *be functors. A natural transformation* $\eta : F \to G$ *is a collection of morphisms*

$$\eta_A : F(A) \to G(A) \qquad A \in Ob(\mathcal{C})$$

such that if $f : A \to B$ *is a morphism in* $\mathcal{C}$*, then*

$$\eta_B \circ F(f) = G(f) \circ \eta_A.$$

3.2 Homological Algebra

Let $\mathcal{A}$ be an abelian category. Let **Ab** be the category of abelian groups.

Lets begin by discussing the notion of Hom. For objects A and B in $\mathcal{A}$, we have an abelian group $\mathrm{Hom}(A, B)$, which is the set of morphism from A to B. This is part of the definition of an abelian category (abelian categories are *pre-additive*). Let us functorialize this in the most obvious way.

Let $f : A \to B$ be a morphism, and let D be an object in $\mathcal{A}$. We transform this into a morphism $\mathrm{Hom}(D, A) \to \mathrm{Hom}(D, B)$ as follows. Let $g : D \to A$ be a morphism. Then $fg : D \to B$ is also a morphism. So (excluding technical details) we get a covariant functor $\mathrm{Hom}(D, -)$ from $\mathcal{A}$ to **Ab**.

When D is in the other slot of Hom, we get a different relationship. Let $g : B \to D$ be a morphism. Then $gf : A \to B \to D$ is also a morphism. This gives us a contravariant functor from $\mathcal{A}$ to **Ab**.

Proposition 26. *For any object* D *in* $\mathcal{A}$*, Hom*$(D, -)$ *is left-exact and co-variant, and Hom*$(-, D)$ *is left-exact and contravariant.*

We say D is *projective* if for every two objects M and N and exact sequence $M \xrightarrow{\varphi} N \to 0$, for every morphism $f : D \to N$, there is a morphism $F : D \to M$ such that the following diagram commutes:

$$\begin{tikzcd} & D \arrow[dl, dashed, "F"'] \arrow[d, "f"] & \\ M \arrow[r, "\varphi"] & N \arrow[r] & 0 \end{tikzcd}$$

We say D is injective when there is an exact sequence $0 \to M \xrightarrow{\varphi} N$, and for every morphism $f : N \to D$ there is a morphism $g : N \to D$ such that the following diagram commutes:

$$\begin{tikzcd} & D & \\ 0 \arrow[r] & M \arrow[u, "f"] \arrow[r, "\varphi"] & N \arrow[ul, dashed, "g"'] \end{tikzcd}$$

Proposition 27. D *is projective if and only if Hom*$(D, -)$ *is exact and injective if and only if Hom*$(-, D)$ *is exact.*

For an object D, we say D is *projective* if $\mathrm{Hom}(D, -)$ is exact and *injective* if $\mathrm{Hom}(-, D)$ is exact. An equivalent characterization of a projective object is as follows. If M and N are objects and $M \xrightarrow{f} N \to 0$ is exact then there is a lift $F : P \to M$ such that the following diagram commutes

Definition 43. *We say a sequence of morphisms* $C_\bullet$

$$0 \to A_0 \xrightarrow{d_1} A_1 \xrightarrow{d_2} \cdots$$

is a cochain complex *if* $d_{n+1}d_n = 0$ *for all* $n \geq 1$. *We define the* n-th cohomology group *to be the quotient* $H^n(C_\bullet) := \ker d_{n+1}/im\ d_n$.

Note that we can take morphisms between cochain complexes (resp, short exact sequences) in an obvious way so we can form $\mathrm{Ch}(\mathcal{A})$, the category of co-chain complexes. This is also an abelian category.

Proposition 28. *If* $0 \to A \to B \to C \to 0$ *is a short-exact sequence in* $\mathcal{A}$, *then we have a long-exact sequence*

$$\cdots \to H^n(A) \to H^n(B) \to H^n(C) \xrightarrow{\delta_n} H^{n+1}(A) \to \cdots$$

Proof. See [Hatcher] or [Dummit&Foote]. This is an application of simultaneous resolution (aka the horseshoe lemma) and the snake-lemma. □

For an object A, a projective resolution is an exact sequencce

$$\cdots P_n \to \cdots \to P_0 \to A \to 0$$

where each P_i is projective. We say $\mathcal{A}$ has *enough projectives* if every object has a projective resolution.

As an example, note that the category of R-modules has enough projectives. Indeed, if M is an R module generated by a set S. Let F_0 be the free module generated by S and let $\epsilon : F_0 \to M$ be the morphism which restricts to the identity on S. Let K_0 be the kernel of ε. This yields a short exact sequence

$$0 \to K_0 \to F_0 \to M \to 0.$$

Then repeat the same process with K_0. Indeed, let T be a generating set of K_0, let F_1 be the free-module generated by T, and let $\varepsilon' : F_1 \to K_0$ be the map which restricts to the identity on T. Then we get an exact sequence

$$0 \to F_1 \to F_0 \to M \to 0$$

which we can continue inductively.

Let us dig deeper into projective resolutions. Let A and D be objects and let $P_\bullet \to A \to 0$ be a projective resolution of A. We can apply $\mathrm{Hom}(-, D)$, which we know is contravariant to get a co-chain complex $0 \to \mathrm{Hom}(A, D) \to \mathrm{Hom}(P_0, D) \to \cdots$. This complex is not necessarily exact, but it is indeed a complex. Hence. we obtain cohomology groups, but we need to prove that these groups are indepdendent of the choice of resolution. We will finish our first discussion of homological algebra with a proof of this fact.

Lemma 6. *Let* $P_\bullet \to A$ *be a projective resolution of* A *and let* $Q_\bullet \to A'$ *be an exact co-chain complex. Let* $f : A \to A'$ *be a morphism. Then for each* $n \geq 0$ *there is a morphism* $f_n : P_n \to Q_n$ *such that the following diagram commutes.*

$$\begin{tikzcd} \cdots \arrow[r] & P_1 \arrow[r] \arrow[d, dashed, "f_1"'] & P_0 \arrow[r] \arrow[d, dashed, "f_0"'] & A \arrow[r] \arrow[d, "f"] & 0 \\ \cdots \arrow[r] & Q_1 \arrow[r] & Q_0 \arrow[r] & A' \arrow[r] & 0 \end{tikzcd}$$

and we say $f_\bullet$ *is a* lift *of* f. *Moreover, any two lifts of* f *are chain-homotopic, i.e. for any other another lift* $g_\bullet : P_\bullet \to Q_\bullet$, *there exists morphisms* $s_n : P_n \to Q_{n+1}$ *such that*

$$d^D_{n+1}s_n + s_{n-1}d^C_n = g_n - f_n$$

and we denote this by $f_\bullet \simeq g_\bullet$.

Proof. As P_0 is projective, and there is a morphism $P_0 \to A \to A'$ there is also morphism from P_0 to Q_0. Call this f_0. Similarly, the map $P_1 \to P_0 \to Q_0$ induces a map $P_1 \to Q_1$ by projectivity. The existence of the other f_n's exist by induction. Finally, we can look at the chain map $f_\bullet - g_\bullet$. We define $s_{-1} = 0$ and obtain a diagonal map $s_0 : P_0 \to Q_1$ by projectivity. The rest can exist by induction. □

3.3 Basic homotopy

Definition 44. *A* path *in* X *is a continuous function* $\gamma : [0,1] \to X$. *We say two paths* γ_1 *and* γ_2 *are homotopic if there is a function* $F : [0,1] \times [0,1] \to X$ *such that* $F(0,t) = \gamma_1(t)$ *and* $F(1,t) = \gamma_2(t)$ *for all* t. *We call such an* F *a* homotopy. *The set* $\pi_1(X, b)$ *defined to be the set of paths* γ *where* $\gamma(0) = \gamma(1) = b$, *modulo homotopy. We can concatenate two paths through* b *in an obvious way. Indeed, if* γ_1 *and* γ_2 *are paths through* b, *then we can define*

$$(\gamma_1 * \gamma_2)(t) = \begin{cases} \gamma_1(2t) & \text{if } 0 \leq t \leq 1/2 \\ \gamma_2(2t) & \text{if } 1/2 < t \leq 1 \end{cases}.$$

Proposition 29. $\pi_1(X, x_0)$ *is a group.*

Proof. Straightforward. □

Proposition 30. *Let $f : X \to Y$ be a continuous map, and fix $x_0 \in X$. Then there is an induced group homomorphism $\pi_1(f, x_0) : \pi_1(X, x_0) \to \pi_1(Y, f(x_0))$*

Proof. Let γ represent a loop in X based at x_0. Then the curve $\eta : [0,1] \to Y$ given by $\eta(t) = f(\gamma(t))$ is a curve in Y based at $f(x_0)$. Moreover, if γ_1 and γ_2 are two curves in X based at x_0 then ... $\square$

Let us compute some examples, where we first give an intutive argument, followed by a rigorous one.

Example 6. *Firstly, consider the unit circle in the complex plane $S^1 \subset \mathbb{C}$, and a basepoint x_0. Basically, loops through x_0 are characterized by their* winding number. *I am using the term a bit loosely here. This characterization is homomorphic. Indeed, if γ_1 and γ_2 have winding numbers r and s, we expect $\gamma_1 * \gamma_2$ to have a winding number of $r + s$. Moreover, we can have negative winding numbers. We are justified by intution to say that $\pi_1(S^1, x_0) \cong \mathbb{Z}$. To prove it, let $\gamma : [0,1] \to S^1$ be a path, and let $\rho : [0,1] \to S^1$ be the function $\rho(t) = e^{2\pi i t}$. Then we can lift γ to a path $\tilde{\gamma} : [0,1] \to \mathbb{R}$, where $\rho(\tilde{\gamma}(t)) = \gamma(t)$ for all t. Then the map $\pi_1(S^1, 1) \to \mathbb{Z}$ given by $\gamma \mapsto \tilde{\gamma}(1)$ is a group isomorphism. So we have $\pi_1(S^1) \cong \mathbb{Z}$.*

Example 7. *As a second example, consider the sphere S^2 and fix a point $p \in S^2$. Then any two loops through p are homotopic to the identity loop $t \mapsto p$. Thus, $\pi_1(S^2) = 0$.*

Lemma 7. *For path connected spaces X and Y and $a \in X, b \in Y$ we have*

$$\pi_1(X \times Y, (a, b)) \cong \pi_1(X, a) \times \pi_1(Y, b)$$

Example 8. *As a final example, let us consider the fundamental group of the torus $\mathbb{T} = S^1 \times S^1$, the donut shape in $\mathbb{R}^3$. Intuitively, we know that there are loops going through the hole, and loops just on the surface - which are indepdendent. Therefore, we are justified in believing that $\pi_1(\mathbb{T})$ has two generators. So we have*

$$\pi_1(\mathbb{T}) = \mathbb{Z} \times \mathbb{Z}$$

as our intuition suggests.

We finish the section with a definition of the homotopy groups $\pi_1, \pi_2, \ldots$
which is consistent with our definition of π_1.

Definition 45. *Let X be a path connected space and let $a \in X$. We define the n-th* homotopy group *$\pi_n(X, a)$ to be the set of continuous functions $S^n \to X$, modulo homotopy, and where the group operation is loop concatentation.*

3.4 (Co)homology

Definition 46. *Define the* standard k-simplex *Δ^k by*

$$\Delta^k = \left\{ (x_0, \ldots, x_k) : x_i \geq 0 \text{ and } \sum x_i \leq 1 \right\}.$$

Remark 8. *There are $k + 1$ extreme points $v_0, \ldots, v_k$ in Δ^k. One can show that $\Delta^k = conv(v_0, \ldots, v_k)$. Moreover, for $0 \leq i \leq k$, the set $F_i = \{(x_1, \ldots, x_k) \in \Delta^k : x_i = 0\}$ is a subset of the boundary $\partial \Delta^k$. Indeeed,*

$$\partial \Delta^k = \bigcup_{i=1}^{k+1} F_i.$$

Definition 47. *A singular simplex is a continuous map $\Delta^k \to \mathbb{R}$.*

Definition 48. *The singular chain group is defined by $C_k(M, \mathbb{R}) = \mathbb{R}$-linear combinations of singular simplicies. We can view this as a real vector space and define the singular cochain group by $C^k(M; \mathbb{R}) = C_k(M; \mathbb{R})^*$ - the dual space of the singular chain group.*

Definition 49. *We define the boundary operator ∂_k by*

$$\partial_n : C_n(M; \mathbb{R}) \to C_{n-1}(M; \mathbb{R}) \qquad \partial_n(\sigma) = \sum_{i=0}^{n} (-1)^i \sigma|_{F_i}.$$

Proposition 31. $\partial_{n-1} \circ \partial_n = 0$ *so* $im\ \partial_n \subset \ker \partial_{n-1}$.

Definition 50. *We consider the chain complex*

$$\cdots \to C_n(M; \mathbb{R}) \to C_{n-1}(M; \mathbb{R}) \to \cdots \to C_1(M; \mathbb{R}) \to \mathbb{R} \to 0$$

and define the singular homology groups *by*

$$H_{n,sing}(M) = \frac{\ker \partial_n}{im\ \partial_{n+1}}.$$

Proposition 32. *(functoriality) If $f : X \to Y$ is continuous then the map*

$$f_* : H_n(X) \to H_n(Y) \qquad f_*(\sigma) = f \circ \sigma : \Delta^n \to Y$$

is an abelian group homomorphism, making

$$H_n : \mathbf{Top} \to \mathbf{Ab}$$

a functor.

Definition 51. *Recall that two maps $f, g : X \to Y$ are* homotopic *if there is a continuous map $F : [0,1] \times X \to Y$ such that $F(0,x) = f(x)$ and $F(1,x) = g(x)$.*

Proposition 33. *(homotopy invariance) If $f, g : X \to Y$ are homotopic then $f_* = g_*$.*

Recall that for a group G, we define its *commutator* $[G,G] = \{xyx^{-1}y^{-1} : x, y \in G\}$, and note that it is a normal subgroup of G. We define the *abelianization* of G by $G^{ab} = G/[G,G]$.

Proposition 34. $H_1(X) \cong \pi_1(X)^{ab}$

Proposition 35. *Let $A \subset X$. Then there is a short exact sequence*

$$0 \to C_\bullet(A) \to C_\bullet(X) \to C_\bullet(X,A) \to 0$$

which induces the following long-exact sequence

$$\cdots \to H_n(A) \to H_n(X) \to H_n(X,A) \xrightarrow{\delta} H_{n-1}(A) \to H_{n-1}(X) \to H_{n-1}(X,A) \xrightarrow{\delta} \cdots$$

Proposition 36. *(Excision) If $U \subset A \subset X$ and $\overline{U} \subset int(A)$ then*

$$H_n(X - U, A - U) \cong H_n(X, A).$$

Corolary 4. *(Mayer-Vietoris) If $X = U \cup V$ then there is a long-exact sequence*

$$\cdots \to H_n(U \cap V) \to H_n(U) \oplus H_n(V) \to H_n(X) \to H_{n-1}(U \cap V) \to \cdots.$$

3.5 De Rham's Theorem

Theorem 5.

$$H^k_{DR}(M) \cong H^k_{sing}(M).$$

We prove this in several steps. First, consider a smooth simplex $\sigma : \Delta^k \to M$ and let $\omega \in \Omega^k(M)$ be a k-form. Recall that $C^k(M;\mathbb{R})$ denotes the cochains on M. Then we can define a co-chain $I(\omega) : C_k(M) \to \mathbb{R}$ by

$$I(\omega)(\sigma) = \int_{\Delta^k} \sigma^*\omega.$$

The goal is to show that the map $I : \Omega^k(M) \to C^k(M)$ induces an isomorphism $H^k_{DR}(M) \cong H^k_{sing}(M)$. Now consider the following diagram.

$$\begin{tikzcd}
\cdots \arrow[r] & \Omega^k(M) \arrow[r, "d"] \arrow[d] & \Omega^{k+1}(M) \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & C^k(M) \arrow[r, "\delta"] & C^{k+1}(M) \arrow[r] & \cdots
\end{tikzcd}$$

We recall that δ is the coboundary operator satisfying $\delta(\phi)(x) = \phi(\partial x)$.

I claim that this diagram commutes. In other words, for $\omega \in \Omega^k(M)$, we have $\delta(I(\omega)) = I(d\omega)$. Indeed,

$$\begin{aligned}
\delta(I(\omega))(\sigma) &= I(\omega)(\partial\sigma) \\
&= I(\omega)(\sum_{i=0}^{k}(-1)^i \sigma|_{[v_0,\ldots,\hat{v_i},\ldots v_k]]}) \\
&= \sum_{i=1}^{n}(-1)^i \int_{F_i} \sigma^*\omega \\
&= \int_{\partial\Delta^k} \sigma^*\omega \\
&= I(d\omega)(\sigma)
\end{aligned}$$

So we do indeed get a well defined map

$$I : H^k_{DR}(M) \to H^k_{sing}(M).$$

I claim that this is an isomorphism.

4 Riemannian Theory

4.1 Basic Definitions

Definition 52. *Let M be a smooth manifold. Let g be a function such that for every $p \in M$, g_p is an inner product on T_pM. If for every $X, Y \in \mathfrak{X}(M)$ the map $M \to \mathbb{R}$ given by $p \mapsto g_p(X_p, Y_p)$ is smooth, then we say g is a* Riemannian metric *on M.*

Remark 9. *We note that an real-valued inner product is symmetric and positive definite by definition.*

Proposition 37. *Every smooth manifold has a Riemannian metric.*

Proof. See [Wikipedia]. □

We can express our metric as

$$g = g_{ij} dx^i \otimes dx^j$$

for some smooth functions $g_{ij} : M \to \mathbb{R}$.

Proposition 38. *The matrix $[g_{ij}]$ is symmetric and positive-definite. Moreover, the functions $g_{ij} : M \to \mathbb{R}$ do not depend on the coordinate chart chosen.*

Proof. Follows from the definitions. □

Definition 53. *For a vector $v \in T_pM$, we define its* length *by*

$$||v|| = \sqrt{\langle v, v\rangle}.$$

Definition 54. *Let $\gamma : [0,1] \to M$ be a piecewise smooth curve. We define its length by*

$$L(\gamma) = \int_0^1 ||\gamma'(t)||dt.$$

Definition 55. *For $p, q \in M$, we can define their distance by*

$$\rho(p,q) := \inf\{L(\gamma) : \gamma : [0,1] \to M \textit{ is a piecewise smooth curve such that } \gamma(0) = p \textit{ and } \gamma(1) = q\}.$$

Proposition 39. *(M, ρ) is a metric space.*

Proof. First, it is clear that $\rho(p,q) \geq 0$ for all $p, q \in M$, and $\rho(p,q) = 0$ if and only if $p = q$. The symmetry property $\rho(p,q) = \rho(q,p)$ is obvious. The only non-obvious part is the triangle inequality, which we prove below.

To show the triangle inequality, let $p, q, r \in M$. Our goal is to show

$$\rho(p,r) \leq \rho(p,q) + \rho(q,r).$$

To this end, define the following paths

α	connecting p, r
β	connecting p, q
γ	connecting q, r
δ	follow β then γ (double speed on both parts)

Then

$$L(\alpha) \leq L(\delta) \leq L(\alpha) + L(\beta).$$

As these paths are arbitrary, this proves that

$$\rho(p,r) \leq \rho(p,q) + \rho(q,r).$$

□

4.2 Connections

Definition 56. *A* connection *on (M, g) is a function*

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$$

such that for every $f, g \in C^\infty(M)$ and $X, Y, Z \in \mathfrak{X}(M)$ and $a \in \mathbb{R}$ we have

1. $\nabla_{fX+gY} Z = f\nabla_X Z + g\nabla_Y Z$ *- $C^\infty(M)$-linearity*
2. $\nabla_X(aY + Z) = a\nabla_X Y + \nabla_X Z$ *- $\mathbb{R}$-linearity*
3. $\nabla_X(fY) = X(f)\nabla_X Y + f\nabla_X Y$ *- a type of Liebniz rule*

Definition 57. *For i, j, we can express $\nabla_{\partial_i}\partial_j = \Gamma^k_{ij}\partial_k$. We call the symbols Γ^k_{ij} the* Christoffel symbols.

Example 9. *Take $M = \mathbb{R}^n$ and use the metric $ds^2 = \sum(dx^i)^2$. Let $X = X^i\partial_i$ and $Y = Y^j\partial_j$ be vector fields. Then we can define the connection by*

$$\nabla_X Y := \sum_{k=1}^n X(Y^k)\partial_k.$$

Therefore, we can express the k-th component of $\nabla_X Y$ as

$$(\nabla_X Y)^k = X(Y^k) = \sum_{i=1}^n X^i \frac{\partial Y^k}{\partial x^i}.$$

In the case where $X = \partial_i$ and $Y = \partial_j$, for any i, j, possibly equal, we have $(\nabla_X Y)^k = 0$ - simply because the coefficients of Y are constants and thus $\dfrac{\partial Y^k}{\partial x^j} = 0$ for all j, k. So $\Gamma^k_{ij} = 0$ for all i, j, k.

Definition 58. *Let (M, g) be a Riemannian manifold and let ∇ be a connection on M. Let $X, Y \in \mathfrak{X}(M)$. Define the* torsion *of X and Y as*

$$T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y].$$

We say the connection ∇ is torsion free *if $T(X,Y) = 0$ for all $X, Y \in \mathfrak{X}(M)$. We say the connection ∇ preserves the metric if for all $X, Y, Z \in \mathfrak{X}(M)$, we have*

$$X(g(Y,Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z).$$

If the connection satisfies these two properties, we say ∇ is a Levi-Cevita connection.

Proposition 40. *(Kozul Formula) A connection ∇ always satisfies*

$$2g(\nabla_X Y, Z) = Xg(Y,Z) + Yg(Z,X) - Zg(X,Y) - g(X,[Y,Z]) - g(Y,[X,Z]) + g(Z,[X,Y]).$$

Corolary 5. *(Levi-Cevita's Theorem) Let (M, g) be a Riemannian manifold. Then there exists a unique torsion-free, metric preserving connection ∇ on M.*

Definition 59. *A smooth curve $\gamma : I \to M$ is a* geodesic *if*

$$\nabla_{\dot\gamma}\dot\gamma = 0.$$

5 Algebraic Varieties

5.1 Commutative Rings and Algebraic Sets

All rings, unless stated otherwise are commutative.

Recall that an *integral domain* or *domain* is a ring where $ab = 0$ implies $a = 0$ or $b = 0$. Moreover, if R is a domain and $a \neq 0$, then if $ab = ac$ then $b = c$, i.e. the *law of cancellation* holds in a domain. This is an equivalent definition. We say that the domain is a *principal ideal domain* or *PID* if every ideal is generated by one element.

As an example $\mathbb{Z}$ is a PID. Indeed, if I is a non-zero ideal, then we can choose elements a and b in I such that $b > a > 1$. Then we can write $b = qa + c$ for some integers q and c satisfying $q \geq 0$ and $0 \leq c < a$. As c is non-negative and a is the smallest positive integer in I, we see that $c = 0$, i.e. $b = qa$. This ring is also a *unique factorization domain*, which is a ring in which each non-zero element decomposes as a unique product of primes.

We call a non-zero element a of R a *unit* if there exists some b such that $ab = 1$. Ideals generated by units are the entire ring. A non-unit which is not the product of any two non-units is called an *irreducible element.* A *unique factorization domain* or *UFD* is a ring in which every non-zero, non-unit element is a unique product of irreducible elements. Hence, $\mathbb{Z}$ is a UFD. We also have the following well-known fact, which is easy to prove.

Theorem 6. *If R is a UFD then so is $R[X]$ and consequently so is $R[X_1, \dots, X_n]$.*

We say R is *Noetherian* if every ideal is finitely generated. We have the following important theorem.

Theorem 7. *If R is Neotherian, then so is $R[X]$ and consequently so is $R[X_1, \dots, X_n]$.*

We use the notation $R[\mathbf{X}]$ instead of $R[X_1, \dots, X_n]$ when it is convenient and it does not cause confusion.

There are two proofs. We give a sketch of both. The first way is to take an ideal I in $R[X]$ and take the set of leading coefficients in I which is a finitely generated ideal in R. We call this ideal J. The idea of the proof is to decompose J into a sum which has a nice pre-image in $R[x]$.

The better way of approaching this is to use the concept of a *Grobner basis.* Unfortunately, I could not find a precise definition, but a Grobner basis is basically a finite generating set of I with "nice algorithmic properties". Thankfully, the exact definition does not matter for us. Basically, if $F = \sum a_i X_1^{i_i} \cdots X_k^{i_k}$, and $G_1, \dots, G_m$ are a set of polynomials in $R[X]$, then we can write $F = \sum A_i G_i + C$, where $\deg C < \deg F$. This can be done by a sort of Euclidean algorithm using lexicographic ordering, called *Bucherberg's Algorithm.*

Given a subset X of R^n, we define its ideal $I(X)$ to be the set of polynomials $f \in R[\mathbf{X}]$ are zero on X. Similarly, for an ideal I of R, we define its *locus* to be the set of points $p \in R^n$ such that $fp = 0$ for all $f \in I$. In general, these maps are not inverses of each other, but we will get to their relationship quite soon. However, we can see some interesting properties. First, we say an algebraic set is *irreducible* if it is not a union of two non-empty algebraic sets, where the two sets in the union are unequal. It is easy to show that V is irreducible if and only if $I(V)$ is prime. Moreover, by Hilbert's basis theorem, we have

Proposition 41. *Every algebraic set is a finite intersection of irreducible algebraic sets.*

Proof. If I is minimally generated by irreducible polynomials $F_1, \dots, F_k$, then $X = Z(F_1) \cap \cdots \cap Z(F_k)$. $\square$

Example 10. $\mathbb{A}^1$ *is irreducible.*

Example 11. *Let $f \in k[x_1, \dots, x_n]$. Then f is irreducible if and only if the ideal (f) is prime, if and only if $Z(f)$ is irreducible.*

Note that if f is reducible if and only if f has a non-trivial factorization $f = gh$. In this case, $Z(g)$ and $Z(h)$ would be proper subsets of $Z(f)$, and we would also have $Z(f) = Z(g) \cup Z(h)$. So we have shown that if f is reducible, then $Z(f)$ can be decomposed as two non-empty algebraic sets.

On the other hand, suppose $Z(f) = X \cup Y$ for non-empty algebraic sets X and Y.

Observation 1. *X is irreducible if and only if $I(X)$ is prime.*

We continue to get more interesting properties of algebraic sets. Suppose $I \subset J$ are ideals. Then $V(J) \subset V(I)$. The situation is simple when k is algebraically closed. If I is a proper ideal of $k[\mathbf{X}]$, then there is some maximal ideal J containing I. But we know J is generated by $\langle x_1 - a_1, \dots, x_n - a_n \rangle$. So $a = (a_1, \dots, a_n) \in V(I)$. Hence

Proposition 42. *If k is algebraically closed and I is a proper ideal of $k[\mathbf{X}]$ then $V(I) \neq 0$.*

Next we define the radical of an ideal I as

$$\operatorname{rad}(I) = \{f \in R[\mathbf{X}] : f^r \in I \text{ for some } r > 0\}.$$

When k is an algebraically closed field, I claim that $I(V(I)) = \operatorname{rad}(I)$. If we had this, then this would allow us to reduce a generating ideal of an algebraic set to a minimal one. Showing that $\operatorname{rad}(I) \subset I(V(I))$ is straightforward. For the other direction, suppose $f \in I(V(I))$. Let $f_1, \dots, f_k$ be the generators of I. We wish to find some r such that $f^r = \sum a_i f_i$ where a_i are members of $k[X_1, \dots, X_n]$. Then the ideal J generated by $f_1, \dots, f_k$ and $1 - x_{n+1} f$ is the entire ring, since if it were not, then $V(J)$ would have at least one element, which it does not, because f vanishes whenever each f_i does. Thus one can write

$$1 = g_1 f_1 + \cdots + g_k f_k + h(1 - x_{n+1} f)$$

for some g_i and h in $k[x_1, \dots, x_{n+1}]$. Substituting $1/f(x_1, \dots, x_n)$ for x_{n+1} results in the equation

$$1 = \sum g_i(x_1, \dots, x_n, \frac{1}{f(x_1, \dots, x_n)}) f_i(x_1, \dots, x_n).$$

We can multiply both sides by a high enough power of f to remove f from the denominator of the rational expression on the right hand side, yielding

$$f^r = \sum c_i f_i$$

for some c_i in I. Thus we have the following theorem

Theorem 8. *(Hilbert's Nullstellensats) $I(V(I)) = \mathit{rad}(I)$.*

This important theorem characterizes gives us a one-to-one correspondence between the radical ideals of $\mathfrak{r}$ and the algebraic sets in $\mathbb{A}^n$. We can go even further. We say $\mathfrak{r}$ is *reduced* if it contains no non-zero nilpotent elements.

Proposition 43. *$\mathfrak{r}/\mathfrak{i}$ is reduced if and only if $\mathfrak{i}$ is a radical ideal.*

Proof. Suppose $\mathfrak{r}/\mathfrak{i}$ is reduced. To show that $\mathfrak{i}$ is radical, suppose $x \in \mathfrak{r}$ is such that $x^n \in \mathfrak{i}$ for some $n \geq 1$. We aim to show that $x \in \mathfrak{i}$. We can safely assume $n > 1$. Then let m be the smallest positive integer such that $2m > n$. Then $y = x^m$ is a nonzero nilpotent element in $\mathfrak{r}$, which is impossible.

Secondly, if $\mathfrak{i}$ is radical, then if $x \in \mathfrak{r}/\mathfrak{i}$ is nilpotent, then $x^2 \in \mathfrak{i}$, so $x \in \mathfrak{i}$. $\square$

Thus we have a one-to-one correspondence between the following sets

$$\text{radical ideals of } \mathfrak{r} \longleftrightarrow \text{ reduced quotients of } \mathfrak{r} \longleftrightarrow \text{ alg. sets}$$

The reader may have noticed, or already knows that the "reduced quotient" is just a ring $k[x_1, \dots, x_n]/I(X)$ where X is an algebraic set. We call this the coordinate ring, and delay a full study to the next section.

Let us sketch out some more properties of algebraic sets before we finish this section. We will finish with giving $\mathbb{A}^n$ a topology, and we will introduce the notion of dimension of algebraic sets.

Proposition 44.

1. *The union of two algebraic sets is algebraic.*
2. *The intersection of two algebraic sets is algebraic.*
3. *$\emptyset$ and $\mathbb{A}^n$ are algebraic.*

Proof. For 1. and 2., we let X and Y be algebraic sets generated by $I_X = (F_1, \dots, F_n)$ and $I_Y = (G_1, \dots, G_m)$. Then note that $X \cup Y$ is generated by the product of the ideals $I_X \cdot I_Y$ and $X \cap Y$ is generated by the sum $I_X + I_Y$. Finally, the ideal of $\emptyset$ is the ring $\mathfrak{r}$ and the ideal of $\mathbb{A}^n$ is $\{0\}$. $\square$

Definition 60. *We give $\mathbb{A}^n$ the* Zariski topology*, which defines the open sets as complements of algebraic sets in $\mathbb{A}^n$. We call open sets* quasi-algebraic sets.

As an example, note that the Zariski-closed sets of $\mathbb{A}^1$ are the finite subsets of $\mathbb{A}^1$. The closed sets in $\mathbb{A}^2$ consist of finite sets, lines and curves.

Definition 61. *An* affine variety *is an irreducible affine algebraic set. A* quasi-affine variety *is a an open subset of an affine variety.*

Remark 1. *The Zariski topology is Hausdorff if and only if k is a finite field...*

We define the *Krull dimension* of a ring to be the supremum of lengths of proper chains of prime ideals, and we denote this by $\dim \mathfrak{r}$. For a prime ideal $\mathfrak{p} \subset \mathfrak{r}$, we define its height to be the supremum of its proper chains of prime ideals.

Proposition 45. *If $\mathfrak{r}$ is Noetherian and $\mathfrak{p}$ is a prime ideal of $\mathfrak{r}$, then*

$$\dim \mathfrak{r} = \dim \mathfrak{r}/\mathfrak{p} + \mathrm{ht}(\mathfrak{p}).$$

Proof. Left to the reader. □

Definition 62. *If X is a topological space, we define its* dimension *to be the supremum of lengths of strictly descending chains of closed sets in X.*

Proposition 46. *Let X be an algebraic set in $\mathbb{A}^n(k)$. Then the topological dimension of X is equal to the Krull dimension of $A(X)$.*

Proof. Let $X \supset Y_0 \supset Y_1 \supset \cdots$ be a descending chain of closed sets in X, where $Y_i \neq Y_{i+1}$. As each Y_i is closed in X, we have $Y_i = Z(\tilde{I}_i)$ where $\tilde{I}_i = I_i/I(X)$ for some prime ideal I_i in $k[x_1, \ldots, x_n]$. We thus get an ascending chain of ideals $I_i \subset \cdots$ which terminates at the Krull dimension of $A(X)$ □

For example, the dimension of $\mathbb{A}^n$ is n.

Proposition 47. *If X is a quasi-affine variety, then* $\dim X = \dim \overline{X}$.

We leave the proof as an exercise to the reader.

Proposition 48. *A variety X in $\mathbb{A}^n$ has dimension $n-1$ if and only if X is generated by a single irreducible polynomial.*

Proof. Suppose X has dimension 1, and let $\mathfrak{p}$ denote its prime ideal. Then by a proposition above, $\mathfrak{p}$ has height 1. By another theorem, this is principle.

Conversely, if X is generated by a principal prime ideal $\mathfrak{p}$, then this ideal has height 1 and proves the theorem. □

5.2 Local Behaviour

We call irreducible affine algebraic sets *affine varieties*. Let V be a variety and let I be its prime ideal. We call $\Gamma(V) = k[X_1, \ldots, X_n]/I$ the *coordinate ring*. Let $\mathcal{F}(V, k)$ be the ring of functions from V into k. Then we say $f \in \mathcal{F}(V, k)$ is a *polynomial function* if there is some $F \in k[x_1, \ldots, x_n]$ such that $f(a) = 0$ if and only if a is a root of F. Note that we get a map $k[x_1, \ldots, x_n] \to \mathcal{F}(V, k)$ defined by $f \mapsto F$ which is a ring homomorphism with kernel I, which shows that we can identify $\Gamma(V)$ with a subring of $\mathcal{F}(V, k)$.

If V and W are varieties in $\mathbb{A}^n$ and $\mathbb{A}^m$, respectively, then a map $\varphi : V \to W$ is called a *polynomial map* if $\varphi(a) = (T_1(a), \ldots, T_m(a))$ for some $T_i \in k[x_1, \ldots, x_n]$. We say it is an isomorphism is there is an inverse polynomial map $\varphi^{-1} : W \to V$. Note that this induces a map $\tilde{\varphi} : \mathcal{F}(V, k) \to \mathcal{F}(W, k)$ given by $\tilde{\varphi}(f)(a) = f(\varphi(a))$. We use the same notation for the induced map $\tilde{\varphi} : \Gamma(W) \to \Gamma(V)$. In this way, we get a one-to-one correspondence between maps $V \to W$ and $\Gamma(W) \to \Gamma(V)$, basically saying that Γ is a contravariant functor.

We denote the field of fractions of $\Gamma(V)$ by $k(V)$, which we interpret as a rational function on V. We say a rational function f is defined at P if it has a representation $f = a/b$ in reduced form such that $b(P) \neq 0$. We denote the set of rational functions defined at P by $\mathcal{O}_P(V)$, which is a subring of $k(V)$. We call this the *local ring at P*. If f is not defined at P, we say P is a pole, and we call the set of poles of f its *poll set*.

Proposition 49. *The poll set of a rational function is algebraic.*

This is nearly trivial, since if $f = a/b$ is a rational function, then its poll set is the locus of b which is algebraic by definition.

Similar to the poll set, we can define which rational functions are defined and vanish at P by

$$\mathfrak{m}_P(V) = \{f \in \mathcal{O}_P(V) : f(P) = 0\}$$

which we call the *maximal ideal of V at P*. Note that this is a maximal ideal as it is the kernel of the evaluation homomorphism $\mathcal{O}_P(V) \to k$.

We say a ring R is local if it contains exactly one maximal ideal. The proofs of the following propositions are straightforward.

Proposition 50. *R is local if and only if its non-units form an ideal.*

Proposition 51. *$\mathcal{O}_P(V)$ is a local ring.*

We define the *tangent space* at P to be the dual module of $\mathfrak{m}/\mathfrak{m}^2$ and we denote this as T_pM. Intuitively, the tangent space is the "linear part" of the functions that vanish at P. We use the dual because we want a morphism $f : X \to Y$ to induce a morphism from T_PX to $T_{f(P)}Y$. If we did not use the dual, the induced morphism would go the wrong way. We call the induced map on the tangent space the *differential* of f denoted by df.

Proposition 52. *The followign are equivalent for an affine curve over k*

1. *f is smooth at P*
2. $\dim_k T_PX = \dim \mathfrak{r}$
3. $f'(P) \neq 0$

Let $\operatorname{Der}(\mathcal{O}_P(X)$ denote the k-algebra of derivations on the ring $\mathcal{O}_P(X)$. If δ is a derivation, then it is a map from $\mathcal{O}_P(X)$ to k. It can be restricted to $m \to k$ and then induced into $\mathfrak{m}/\mathfrak{m}^2 \to k$. Essentially, we have an isomorphism $\operatorname{Der}(\mathcal{O}_P(X) \cong (\mathfrak{m}/\mathfrak{m}^2)^*$.

We have $\mathcal{O}_P(X) = \mathcal{O}_P(\mathbb{A}^n)/I$, where I is the ideal of X. Also, we are comfortable with the fact that

$$\mathfrak{m}/\mathfrak{m}^2$$

are just the functions of degree ≤ 1 which vanish at P, i.e. it is the set of functions

$$(x, y) \mapsto ax + by.$$

Say I is generated by F. Then there is a tangent line given by

$$x\frac{\partial F}{\partial x}(P) + y\frac{\partial F}{\partial y}(P) = 0.$$

This equation must hold in $\mathfrak{m}/\mathfrak{m}^2$.

Notice that each $x_i\dfrac{\partial F}{\partial x_i}(P)$ is a member of $T_PX = \mathfrak{m}/\mathfrak{m}^2$. Moreover, linear and polynomial combinations of these elements are also members of T_PX. Recall that $\mathfrak{m} = (x_1 - a_1, \ldots, x_n - a_n)$.

5.3 Projective algebraic sets

We define *projective space* $\mathbb{P}^n$ to be the equivalence classes in $\mathbb{A}^n - \mathbf{0}$ where $(a_1, \ldots, a_n) \sim (b_1, \ldots, b_n)$ if and only if there exists some $\lambda \neq 0$ such that $a_i = \lambda b_i$.

With affine varieties, we looked at the zero sets of sets of polynomials. These corresponded to radical ideals. It should be no surprise that in the projective case, we consider the zero sets of homogeneous polynomials. Indeed, these are the polynomials that are defined on projective space. For a set of homogeneous polynomials T, we define

$$Z(T) = \{P \in \mathbb{P}^n : f(P) = 0 \text{ for all } f \in T\}.$$

We call such sets *projective algebraic sets.* Then the following theorem is a direct consequence of Hilbert's Nullstellensats.

Theorem 9. *There is a one-to-one correspondence between projective algebraic sets and homogeneous radical varieties.*

Now we equip $\mathbb{P}^n$ with the Zariski topology by defining the open sets of $\mathbb{P}^n$ to be complements of algebraic sets. Note that on $\mathfrak{r} = k[x_1, \ldots, x_n]$, we have coordinate functions x_i. Let $H_i = Z(x_i)$ and let $U_i = \mathbb{P}^n - H_i$, which is open. Note that $\mathbb{P}^n$ is covered by $U_1, \ldots, U_n$. Indeed, if $u = (u_1, \ldots, u_n) \in \mathbb{P}^n$ then at least one u_i is non-zero and hence $u \in U_i$. Then for each i, define $\varphi_i : U_i \to \mathbb{A}^n$ by

$$\varphi(u_1, \ldots, u_n) = \left(\frac{u_1}{u_i}, \ldots, \frac{u_n}{u_i}\right).$$

Proposition 53. *Each φ_i is a homeomorphism from U_i onto $\mathbb{A}^n$.*

Proof. Note that $\varphi = \varphi_i$ is bijective. □

In turn, this tells us about the structure of a variety in $\mathbb{P}^n$.

Proposition 54. *Let X be a projective variety. Then X is covered by $X \cap U_i$ and each $X \cap U_i$ is homeomorphic to a variety in $\mathbb{A}^n$.*

Note that, informally speaking, this makes $\mathbb{P}^n$ a "manifold" with "atlas" (φ_i, U_i).

Example 12. *Two isomorphic projective varieties that have non-isomorphic coordinate rings.*

$f_1 = x_0x_2 - x_1^2$ *and* $f_2 = x_0x_2 - x_1^2$.

5.4 Varieties

5.5 Elliptic Curves

6 Fluid Dynamics

6.1 Basic Definitions

6.2 The Continuity Equation

Consider a barell filled with water. There is an initial force which moves the water, and it keeps moving for some time. At each time t, there is a velocity field X_t of the fluid.

Definition 63. *(Fluid system) Let $M \subset \mathbb{R}^3$ be a smooth oriented 3-dimensional Riemannian manifold with boundary, and let Vol be its volume form. Let $X_t \in \Gamma(TM)$ denote its velocity field, and let $\Phi_{t,s}$ denote its 2-parameter flow. In other words, for a fixed t_0, $\Phi_{t_0,s}$ is the flow of X_{t_0}, so*

$$\frac{d}{dt}\Phi_{t,s}(p) = X(t, \Phi_{t,s}(p)), \quad \Phi_{s,s}(p) = p.$$

Let Ω_0 be a connected, compact 3-dimensional submanifold, and let $\Omega_t = \Phi_{t,0}(\Omega_0)$ be its image along the flow. Let $\rho : M \times \mathbb{R} \to [0, \infty)$ be the mass-density field of M, which we assume is C^1. We call (M, X_t, Ω_0, ρ) a fluid system*.*

Remark 10. *To simplify things, we use the notation $\rho_t = \dfrac{\partial \rho}{\partial t}$ and we let u denote the velocity field. We also define Define*

$$m(t) = \int_{\Omega_t} \rho dV.$$

Remark 11. *Φ_t is a diffeomorphism of M. Therefore,*

Theorem 10. *(Reynold's transport theorem) Let (M, X_t, Ω_0, ρ) be a fluid system. Define*

$$m(t) = \int_{\Omega_t} \rho dV.$$

Then

$$m'(t) = \int_{\Omega_t} (\rho_t + \nabla \cdot (\rho u))dV.$$

Corolary 6. *(Continuity equation) By Reynold's transport theorem, if we assume the mass is constant through time, i.e. $m' = 0$, then we get the continuity equation*

$$\rho_t + \nabla \cdot (\rho u) = 0.$$